[19]:
```
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import BaggingClassifier,AdaBoostClassifier
from sklearn.metrics import accuracy_score,classification_report,confusion_matrix

d=load_breast_cancer()
x=pd.DataFrame(d.data,columns=d.feature_names)
y=d.target
print("DATASET SHAPE:",x.shape)

x_train,x_test,y_train,y_test=train_test_split(x,y,stratify=y,random_state=42,test_size=0.3)

bag=BaggingClassifier(estimator=DecisionTreeClassifier(),n_estimators=50,random_state=42)
bag.fit(x_train,y_train)
p1=bag.predict(x_test)

boost=AdaBoostClassifier(estimator=DecisionTreeClassifier(max_depth=1),n_estimators=50,random_state=42)
boost.fit(x_train,y_train)
p2=boost.predict(x_test)

for name,pred in [("BAGGING",p1),("BOOSTING",p2)]:
    print("\n",name)
    print("ACCURACY SCORE\n",accuracy_score(y_test,pred))
    print("CONFUSION MATRIX\n",confusion_matrix(y_test,pred))
    print("CLASSIFICATION REPORT\n",classification_report(y_test,pred))
```

```
DATASET SHAPE: (569, 30)

 BAGGING
ACCURACY SCORE
 0.9415204678362573
CONFUSION MATRIX
 [[ 58   6]
 [  4 103]]
CLASSIFICATION REPORT
              precision    recall  f1-score   support

           0       0.94      0.91      0.92        64
           1       0.94      0.96      0.95       107

    accuracy                           0.94       171
   macro avg       0.94      0.93      0.94       171
weighted avg       0.94      0.94      0.94       171


 BOOSTING
ACCURACY SCORE
 0.9590643274853801
CONFUSION MATRIX
 [[ 58   6]
 [  1 106]]
CLASSIFICATION REPORT
              precision    recall  f1-score   support

           0       0.98      0.91      0.94        64
           1       0.95      0.99      0.97       107

    accuracy                           0.96       171
   macro avg       0.96      0.95      0.96       171
weighted avg       0.96      0.96      0.96       171
```

[20]:
```
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import BaggingClassifier,AdaBoostClassifier
from sklearn.metrics import accuracy_score,classification_report,confusion_matrix
```

[21]:
```
d=load_breast_cancer()
x=pd.DataFrame(d.data,columns=d.feature_names)
y=d.target
print("DATASET SHAPE:",x.shape)
```

```
DATASET SHAPE: (569, 30)
```

[22]:
```
x_train,x_test,y_train,y_test=train_test_split(x,y,stratify=y,random_state=42,test_size=0.3)
```

[23]:
```
bag=BaggingClassifier(estimator=DecisionTreeClassifier(),n_estimators=50,random_state=42)
bag.fit(x_train,y_train)
p1=bag.predict(x_test)
```

[24]:
```
boost=AdaBoostClassifier(estimator=DecisionTreeClassifier(max_depth=1),n_estimators=50,random_state=42)
boost.fit(x_train,y_train)
p2=boost.predict(x_test)
```

[25]:
```
for name,pred in [("BAGGING",p1),("BOOSTING",p2)]:
    print("\n",name)
    print("ACCURACY SCORE\n",accuracy_score(y_test,pred))
    print("CONFUSION MATRIX\n",confusion_matrix(y_test,pred))
    print("CLASSIFICATION REPORT\n",classification_report(y_test,pred))
```

```
 BAGGING
ACCURACY SCORE
 0.9415204678362573
CONFUSION MATRIX
 [[ 58   6]
 [  4 103]]
CLASSIFICATION REPORT
              precision    recall  f1-score   support

           0       0.94      0.91      0.92        64
           1       0.94      0.96      0.95       107

    accuracy                           0.94       171
   macro avg       0.94      0.93      0.94       171
weighted avg       0.94      0.94      0.94       171


 BOOSTING
ACCURACY SCORE
 0.9590643274853801
CONFUSION MATRIX
 [[ 58   6]
 [  1 106]]
CLASSIFICATION REPORT
              precision    recall  f1-score   support

           0       0.98      0.91      0.94        64
           1       0.95      0.99      0.97       107

    accuracy                           0.96       171
   macro avg       0.96      0.95      0.96       171
weighted avg       0.96      0.96      0.96       171
```

[ ]: